National PG College, Lucknow – 226001
(An Autonomous College of University of Lucknow)
www.npgc.in

Bachelor of Computer Application (BCA)

NEP

2023-2024

SEMESTER	PAPER NO.	PAPER NAME
BCA I	101	Computer Fundamentals and Emerging Technologies
	102	Discrete Mathematical Structure
	103	Digital Electronics
	104	System Analysis and Design
	105	Programming in 'C' Language
	106	Practical[C]+Minor Project in C
	Minor	Information Technology
	Vocational	*Office Automation I (MS Word and MS Powerpoint)*
BCA II	201	Data Structures
	202	Oral And Written Communication Skill
	203	Computer Networks
	204	Programming in C++
	205	Operating System
	206	Practical[Data Structures, C++]
	Vocational	*Office Automation II (MS Excel and MS Access)*
BCA III	301	Database Management System (Using SQL Server)
	302	Programming in PYTHON
	303	Computer Architecture
	304	Designing & Analysis of Algorithms
	305	Organisational Behavior
	306	Practical [DBMS, Python]
	Minor	Principles of Organisational Theories
	Vocational	*ICT Tools and its Applications*
BCA IV	401	Cloud Computing
	402	Java Programming
	403	Software Engineering
	404	Web Designing
	405	Computer Oriented Numerical Methods
	406	Practical (Java, HTML,CSS, JAVASCRIPT), Mini Project
	Vocational	*Introduction to Web Designing*
BCA V	501	Programming & Problem Solving Through C#
	502	Mobile Application Development Using Flutter
	503	Artificial Intelligence and Data Analytics
	504	Software Testing and Quality Management
	505	Computer Graphics
	506	Practical[C#]
	Qualifying	*Summer Internship*
BCA VI	601	Data Warehousing and Mining
	602	Linux & Shell Programming
	603	Cryptography and Network Security
	604	Industrial Training and Project
	Qualifying	*Research Presentation/Publication*

SEMESTER – I

Paper No.	**101**
Paper Name	**Computer Fundamentals and Emerging Technologies**

UNIT -I

Introduction, Characteristics of Computers, Block diagram of computer,Types of computers and features: Mini Computers, Micro Computers, Mainframe Computers, Super Computers(03).Types of Programming Languages: Machine Languages, Assembly Languages, High Level Languages(03), Translators:- Assembler ,Compiler, Interpreter and Linkers(02),Operating system concepts, Types of OS, Functions of OS (04).

Lectures: 12

UNIT -II

The Evolution of a computer program, computer program structure: the structured control flow constructs of selection (if/then/else) and repetition (while and for), block structures, and subroutines. (03) Memory concepts, Types of Memory (Primary and Secondary):-RAM,ROM and its types, Secondary Storage Devices (01) , Types of variables: local variable, global variable, static variable, automatic variable, external variable, computer program functionality, how data processing works, standard I/O operations(03), Data Organization: Drives, Files, Directories(03).

Lectures: 10

UNIT -III

Basic elements of a communication system, Data transmission modes, Data Transmission speed, Data transmission media (04),Digital and Analog Transmission, Network topologies, Network Types (LAN, WAN and MAN), Client and Servers , Intranet, Extranet.(04) Internet: Introduction to Internet, Terminologies related to Internet: Protocol, Domain name, IP address, URL, World Wide Web, Connecting to the Internet,(04)

Lectures: 12

UNIT -IV

Introduction to Big Data: Types of digital data, Big Data architecture and characteristics, Big Data technology components, Big Data importance and applications. (05) Introduction to Block Chain: Structure, Operational aspects of Bitcoin Block, Compare different types of Block chains, Protocols, Payment mode code execution.(06)

Lectures: 11

Total Lectures:45

SEMESTER – I

Paper No.	**102**
Paper Name	**Discrete Mathematical Structures**

UNIT -I

Sets: Sets, Subsets, Equal Sets Universal Sets, Finite and Infinite Sets, Operation on Sets, Union, Intersection and Complements of Sets, Cartesian product, Cardinality of Set, Simple Applications. (06)Relations And Functions**:** Properties of Relations, Equivalence Relation, Partial Order Relation Function: Domain and Range, Onto, Into and One to One Functions, Composite and Inverse Functions, Introduction of Trigonometric, Logarithmic and Exponential Functions.(06)

Lectures: 12

UNIT -II

Partial order relations and lattices: Partial Order Sets, Representation of POSETS using Hasse diagram, Chains, Maximal and Minimal Point, Glb, lub(06), Lattices & Algebraic Systems, Principle of Duality, Basic Properties, Sub lattices, Distributed & Complemented Lattices(06)

Lectures: 12

UNIT -III

Number Theory: Greatest Common Divisors, Euclidean Algorithms, Fibonacci Numbers, Complexity of Euclidean Algorithms, Congruences and Equivalence Relations (04) Groups & Sub groups: Group axioms, permutation groups, subgroups, cosets, normal subgroups, semi-groups, free semi-groups, 89monoids (04), error correcting codes (02).

Lectures: 10

UNIT -IV

Combinatorics & Recurrence Relations: Basic Theorems on permutation and combinations. Pigeon hole principle, principle of inclusion and, exclusion (04). Ordinary & exponential generating functions (03) recurrence relation, solving recurrence relation by substitution, solving recurrence relation by conversion to linear recurrence relation (04)

Lectures: 11

Total Lectures:45

SEMESTER – I

Paper No.	**103**
Paper Name	**Digital Electronics**

UNIT -I

Binary System : Digital Computers and digital systems , Number system(03), Binary Arithmetic , Signed binary numbers, complements : r's complements(2's complement, 10's complements), (r-1)'s complements (1's complement, ,9's complement) , Subtraction using 1's complement and 2's complement method , Binary codes ,Logic gates: Inverter, AND, OR, NOR, NAND, XOR, XNOR, De-Morgan's Theorems, Boolean algebra, Canonical and standard forms , Karnaugh Map (04) , Introduction to IC Digital logic families(RTL, DTL, TTL, ECL, MOS and CMOS) and characteristics (Fan-In ,Fan-out, Power Dissipation , Propogation delay ,Noise margin) of IC Digital logic families (TTL , ECL , CMOS) (03)

Lectures: 10

UNIT -II

Combinational and sequential Circuits : Half adder , Full adder , Half Subtractor , full Subtractor , Serial and parallel adder (04) , Code conversion circuits(02) , Parity generator and checker (02) , Comparators(01) , Encoder , Decoder , Multiplexer , De-multiplexer (03)

Lectures: 12

UNIT -III

Flip flops: RS-Flip flop, D-Flip flop, T-flip flop, JK-Flip flop , JK Master slave flip flop, edge triggered and pulse triggered flip flops (06), Registers: Simple Register , Shift registers ,Types of Shift Register , Construction and timing diagram of 4-bit Shift Register (06).

Lectures: 12

UNIT -IV

Counter - Binary counter, characteristic of counter-(Synchronous/Asynchronous, Modulus/length of counter, UP/DOWN counter, Speed of Asynchronous counter) Construction of full length and Non full length, UP/DOWN Synchronous counter using 2, 3 and 4 FF, Ripple counter. Design of different MOD counter, Construction of Asynchronous counter, Ring Counter, Johnson Counter(05) . Random Access Memory, Read Only Memory, Types of Read Only Memory(03), Multivibrators- Astable, Monostable, Bistable, Schmitt Trigger, Timer (03)

Lectures: 11

Total Lectures: 45

SEMESTER – I

Paper No.	**104**
Paper Name	**System Analysis and Design**

UNIT -I

System concept, Definition, Characteristics of system, Elements of a System: Input, Output Processor, Control, Feedback, Environment, Boundaries and Interface, Principles of System, (04) **Types of System:**-Abstract and Physical system, Open and Closed System, Deterministic and Probabilistic System, Need of Information, Types of Information, Man made Information Systems, : DSS, MIS, TPS.(05) **system analyst** ,role of system analyst, qualification and responsibilities of System Analyst, Multifaceted role of System analyst(03).

Lectures:11

UNIT -II

System development life cycle and its various phases:-Preliminary investigation, determination of system requirements, Development of software, System testing, Implementation, evaluation and maintenances, System documentation: Types of documentation and their importance. (05) General structure of SRS**.** System Flow chart and its symbols,**Software Crisis**: From programmers' point of view, from users' point of view, System Panning, Feasibility study: steps in feasibility study, its report and importance (05)

Lectures:10

UNIT -III

System Analysis - Information Gathering and its various Tools and techniques, tools of Structured Analysis:- (02) ER diagram, data flow diagrams, Data Dictionary, Decision Tree, Structured English, Decision Table (06), Introduction of Control Flow Graph, CPM, PERT chart, Gantt chart.(02) System Design Concepts: Module specifications, Module Coupling and cohesion, Top-down and bottom-up design; Logical and Physical design,(02)

Lectures:12

UNIT -IV

System design - Input design: Input data, Input media and devices; Output design; Form Design: Classification of forms, Requirements of Form design. **data base design**:-objectives of database,Types of relationship, types of data structure, (04) **system testing and quality assurance** -reason of system testing, nature of test data, Test plan ,types of System test ,Goal of Quality assurance in SDLC, levels of Quality Assurance(05) , Introduction of implementation: System Conversion, Conversion methods, User Training, primary activities of a maintenance procedure and types of maintenance(03).

Lectures:12

Total Lectures:45

SEMESTER – I

Paper No.	**105**
Paper Name	**Programming in 'C' Language**

UNIT-I

Basic Programming Tools: Flowchart and Algorithm, Structured and Procedural Programming Concepts, History and Features of C, General Structure of C program, C character set, C Tokens: Keywords, Identifiers, Literals/Constants, Variable concept, Expression and statement, Various Data types in C (05), enum, Operators and expression: Arithmetic, Relational, assignment, logical, bitwise, sizeof operator, comma operator, ternary operators. Basic I/O functions. Compilation process of C program, Header files, Library functions, Preprocessor directives: #include, #define, (05).

Lectures: 11

UNIT-II

Control Statements: If Statement, Switch Statement, Unconditional Branching using goto statement, Construction of loops and implementation While Loop, Do While Loop, For Loop, Break and Continue statements (06) nested loops, Pointer: Declaration, referencing and dereferencing pointer, operations on pointers. storage classes: automatic, register, external and static (05)

Lectures: 11

UNIT-III

Introduction to Arrays: Declaration, Initialization One dimensional array Two dimensional arrays, Matrix Operations. Bubble sort, Selection sort, Insertion Sort. Linear and binary Searching, array of pointers. (06) Strings, String operations: length, compare, concatenate, copy etc. Functions: functions prototypes, types of arguments, function call, call by value, call by reference, Recursion, array and function. (07)

Lectures: 13

UNIT-IV

Structure Declaration, Initialization, Structure variable, Structure Assignment, Nested Structures, Structures and Functions, Structures and Arrays, typedef, Unions Declarations, (06) File handling: Types of file processing: Sequential access, Random access Open, Close, Create, Process, unformatted data file. Command line arguments. (06)

Lectures: 10

Total Lectures: 45

SEMESTER – I

Paper No.	**106**
Paper Name	**Practical [C] + Minor Project in C**

SEMESTER – I

[Minor Paper]

Paper Code	
Paper Name	**Information Technology [Minor]**

UNIT-I

Introduction, Characteristics of Computers, Block diagram of computer, Types of computers and features: Mini Computers, Micro Computers, Mainframe Computers, Super Computers. Types of Programming Languages: Machine Languages, Assembly Languages, High Level Languages, Translators: - Assembler, Compiler, Interpreter and Linkers, Operating system concepts, Types of OS, Functions of OS

Lectures: 15

UNIT-II

I/O Devices: - Keyboard, Mouse, Scanner, Light pen, Trackball, Joystick, Bar NUMBER reader, OCR, OMR, MICR, Digitizer; Monitor, Printer, Plotter etc., Memory concepts, Types of Memory (Primary and Secondary):-RAM,ROM and its types, Secondary Storage Devices (Magnetic tape, Magnetic Disk(FD, HD),Optical Disk(CD,DVD), Pen drive. Data Organization: Drives, Files, Directories

Lectures: 15

UNIT-III

Basic elements of a communication system, Data transmission modes, Data Transmission speed, Data transmission media, Digital and Analog Transmission, Network topologies, Network Types (LAN, WAN and MAN), Client and Servers, Intranet, Extranet. **Internet:** Introduction to Internet, Terminologies related to Internet: Protocol, Domain name, IP address, URL, World Wide Web, Connecting to the Internet.

Lectures: 15

UNIT-IV

Introduction to Big Data: Types of digital data, Big Data architecture and characteristics, Big Data technology components, Big Data importance and applications, Introduction to Block Chain: Structure, Operational aspects of Bitcoin Block, Compare different types of Block chains, Protocols, Payment mode code execution.

Lectures: 15

Total Lectures: 60

SEMESTER- I

[Vocational Paper]

Paper Code	
Paper Name	**Office Automation I (MS Word and MS Powerpoint)**

SEMESTER – II

Paper No.	**201**
Paper Name	**Data Structures**

UNIT-I

Data Structure, definition, and application, Arrays: Representation of single and multidimensional arrays; sparse arrays - lower and upper Triangular matrices and Tri-diagonal matrices, Recursion, Defining Stack and Queue, Stack Operations and Implementation, Array Implementation, Pointer Implementation,(05) Stack Applications, Convert Number Bases by Using Stacks, Infix to Postfix Conversion, Evaluation of postfix expression. Introduction and primitive operation on queues, D-queues Queue Application, Circular, Double Ended and Priority Queues.(05)

Lectures: 10

UNIT-II

Lists, Basic Terminology, Static Implementation of Lists, Pointer Implementation of Lists, Insertion in a list, Deletion from a list, traversal, Searching,(05) Arrays using Linked List, Doubly Linked Lists, Circular, Doubly, Circular Doubly Linked List(07)

Lectures: 12

UNIT-III

Defining Graph, Basic Terminology, Graph Traversal: Depth First Search (DFS), Breadth First Search (BFS), Shortest Path Problem,(03) Trees: Introduction and terminology; Traversal of binary trees; Algorithms for tree operations such as traversal, insertion, deletion; Binary Trees, Inorder, Postorder and Preorder Traversals,(03) Minimal Spanning Tree, Prims and Kruskals Algorithm, Binary Search Trees, Operations on a BST, Insertion, Deletion , Search for a key in BST, B-Trees: Introduction: Threaded Binary tree, AVL Trees(06)

Lectures: 12

UNIT-IV

Searching and Sorting techniques, Sequential Search, Binary Search, Selection Sort, Insertion Sort, Bubble Sort(05), Quick Sort, 2-way Merge Sort, Heap Sort, Bucket or Radix sort. Hashing.(06)

Lectures: 11

Total lectures:45

SEMESTER – II

Paper No.	**202**
Paper Name	**Oral And Written Communication Skill**

UNIT-I

Introduction to Communication : Meaning and process of communication. Scope of communication Channels of communication. (04) Models of communication – SMCR Model, Shannon-Weaver Model Barriers to communication, Verbal communication, 7 C's of communication, Meta Communication.(06)

Lectures: 10

UNIT-II

English Grammar and Written Communication English Grammar – Tense/Article, Preposition, Modals, Transformation of sentences. Written communication – principles, merits & demerits. Business letters – Structure/Parts of a letter, essentials of effective business letters. (06) Sales letters – qualities, using effective sales statements, Types of letters – sales, complaints and adjustments, placing orders (06)

Lectures: 12

UNIT-III

Oral Communication Oral communication – advantages and disadvantages. Interviews – meaning and purpose, Types of interview. Guidelines for interviewer and interviewee. (06) Important non-verbal aspects in an interview - Proxemics, Paralanguage. Styles of interviewing, Cover letter, Job application, Resume-writing. (06)

Lectures: 12

UNIT-IV

Reading and Listening Reading – Meaning and importance in communication, Listening – meaning, nature and importance, Barriers to listening (05) Kinesics – Role of body language in communication Group discussion – methodology, guidelines, role and functions, Practice role playing and group discussions (06)

Lectures: 11

Total Lectures: 45

SEMESTER – II

Paper No.	**203**
Paper Name	**Computer Networks**

UNIT - I

Computer networks, Networks Hardware-----Local Area network, Metropolitan Area network, Wide Area network, (03) Wireless network, Internetworks, (03) Networks Software: Protocol Hierarchies, Design and Issues for layers, (02) Interfaces and Services, Connection oriented and Connection less Services (03)

Lectures: 11

UNIT- II

Reference Models, and OSI Reference Model, TCP/IP Protocol Suit (03) The Physical Layer: Maximum Data Rate of a channel, Transmission Media: Magnetic Media, Twisted Pair, Baseband and Broadband Coaxial cable, Fiber Optical (03) Wire less Transmission, structure of telephone system, Switching, Multiplexing, ISDN (04)

Lectures: 10

UNIT -III

Data link layer, Error control, Flow control, Sliding Window Protocol, (04) Channel Allocation Problem, Multiple Access Protocol: ALOHA, CSMA protocol, Collision Free protocol, Polling, (04) Network layer: Routing Algorithm, Congestion Control Algorithm, IP protocol, IP Addresses. (04)

Lectures: 12

UNIT -IV

Transport Layer: Addressing, Establishing and releasing a connection, (04) TCP service Model, TCP protocol (04), the Application Layer: Network Security, Domain Name System(04)

Lectures: 12

Total Lectures: 45

SEMESTER – II

Paper No.	**204**
Paper Name	**Programming in C++**

UNIT-I

OOP concept, Procedural vs OOP Programming, OOP terminology and features (03), Tokens, Character set, Keywords, Data-types Declarations, Constants and variables, Expressions , Standard Library and header files(03) Operator and Expressions; Arithmetic Operator, Increment/Decrement Operator, Relational Operator, Logical Operator and Conditional Operators, Logical Expression, Flow of control statement; Selection statement, Iteration Statement, Jump statement, Construction of loops and implementation , while, Do-while, for statements nested loops.(04).

Lectures:10

UNIT-II

Arrays one dimensional and two arrays.(01) Classes and Objects: Need for Classes, Encapsulation, Information Hiding (03), Declaration of Classes, iostream operators, referencing class Members,(04) Scope of Class and Its members, Scope resolution operator and its uses, Nested Classes(04)

Lectures:12

UNIT-III

Functions in class: function definition, Default arguments, Constants arguments, Call by value, Call by reference, Calling Functions with arrays, returning from a function,(04) storage class specifies automatic, external and static variable (03), Function overriding, Function overloading, Operator Overloading. Functions: Friend, Inline, Abstract, Virtual, Pure Virtual.(05)

Lectures:12

UNIT-IV

Constructors and Destructor: Declaration, Definition and characteristics (03) Default Constructor, Copy constructor, Parameterized constructor, constructor overloading (04) Inheritance: Need, Different forms, single Inheritance, Multiple Inheritance. (04)

Lectures:11

Total Lectures:45

SEMESTER – II

Paper No.	**205**
Paper Name	**Operating System**

UNIT-I

Introduction to OS, Objectives of OS, Functions of OS - Process Management, Device Management, Memory Management, Information Management (04) Types of operating systems: Batch System, Multiprogramming System, Timesharing System, Multiprocessing System, Parallel System, Distributed Operating System, Real time System and Embedded System (06)

Lectures: 10

UNIT-II

Process Management: Process Concept, Life Cycle of Process, Process Schedulers, CPU Scheduling Criteria, Preemptive and Non-Preemptive Scheduling, Scheduling Algorithms: FCFS, SJF, SRTF, RR, Priority Scheduling (06) Dead Lock: The Critical-Section Problem, Semaphores, Resource allocation graph, Introduction: Characterization, Deadlock Necessary conditions, Deadlock prevention, Deadlock Avoidance (Banker's Algorithm), Safe state condition, Deadlock Detection and Recovery. (06)

Lectures: 12

UNIT-III

Memory Management: Characteristics, Logical and Physical Address space, Swapping, Fixed Partitioned and Variable Partitioned Memory Models, Fragmentation: Internal and External (06) Placement Algorithms: First Fit, Next Fit, Best Fit, Worst Fit. Virtual Memory Concept, Demand paging, Logical to Physical Mapping, Page Fault and Page replacement algorithms: FIFO, LRU, OPR (06)

Lectures: 12

UNIT-IV

File System: File concept, File operations, Types of Files, File Access methods, Directory Structure, File-System Implementation: File system structures, Directory Implementation, Allocation methods: Contiguous, Linked, Indexed (06) Disk Management: Disk Structure & Scheduling Methods, Free Space Management, Disk management Algorithms FCFS, SSTF, SCAN, LOOK, C-SCAN, C-LOOK (05)
Lectures: 11

Total Lectures:45

SEMESTER – II

Paper No.	**206**
Paper Name	**Practical (Data Structure, C++)**

SEMESTER – II

[Vocational Paper]

Paper No.	
Paper Name	**Office Automation II (MS Excel and MS Access)**

SEMESTER – III

Paper No.	**301**
Paper Name	**Database Management System (Using SQL Server)**

UNIT -I

Data, Information and knowledge, introducing database and different kinds of database users, concept or a database, interacting with a database,(03) architecture of a database, using relational databases, basics of relational databases, using relational databases(03), identifiers for relations, characteristic of database, database system concepts and data independence, content of data dictionary, data administration function.(04)

Lectures: 10

UNIT -II

Traditional data model – ANSI/SPRC, 3-level architecture, over view, of three traditional models – hierarchical, network and relational models, comparison of these models(02), ER model(02), File organization technique – random, file organization technique, multi key file organization technique, Indexing (03), concurrency control, database security, database recovery(04)

Lectures:11

UNIT -III

Data Normalization (02) Introduction to SQL: Characteristics and advantages, SQL Data Types and Literals, DDL, DML, SQL Operators (04), Tables: Creating, Modifying, Deleting, Views: Creating, Dropping, Updating using Views SQL DML Queries: SELECT Query and clauses, Set Operations, Predicates and Joins,.Aggregate Functions, Nested Queries(06)

Lectures:12

UNIT -IV

Specifying constraints and indexes in SQL, data manipulation, multiple table operations (06) Database Modification using SQL Insert, Update and Delete, T-SQL: Cursor, concept of Stored Procedures, Functions and Triggers (06)

Total Lectures:12

Total Lectures:45

SEMESTER – III

Paper No.	**302**
Paper Name	**Programming in Python**

UNIT-I

Introduction to Python: Features, Installation, setting up path and Working with Python Understanding Python variables, Python basic Operators, Understanding python blocks, Python Data Types: Numeric, Boolean, None, String, List, Tuple, Dictionary. Python Program Flow Control Conditional blocks using if, else and nested if-else. (05) Loops in python: simple loops and for loop using range in string, list, tuple and dictionaries. Use of while loops in python, Nested loop, Loop manipulation using pass, continue, break and else. String Manipulation: Accessing Strings, Basic Operations, String slicing.(05)

Lectures: 10

UNIT-II

Lists: Introduction, accessing list, Operations, working with lists and methods, Tuples: Accessing Tuples, Operations, working and methods. Dictionaries: Accessing values in Dictionaries, working with dictionaries, properties, functions. List Comprehensions, Dictionary Comprehension. (05) Functions: Defining a function, calling a function, Types of functions: Regular and Anonymous functions (Lambda Functions), Function arguments and its types, Global and Local variables.(06)

Lectures: 11

UNIT-III

Modules: Importing modules, Math module, Random module. File Handling: opening and closing files, reading and writing files, os module. (04) Exception Handling: Types of Exceptions, User Defined Exceptions, Raising an Exception. (03) OOPs concepts: Class and object, Built-in Class Attributes, Operator Overloading, Method Overloading and overriding, concept of Inheritance.(05)

Lectures: 12

UNIT-IV

NumPy - Basics, creating arrays, Slicing, Matrix Operations. Plotting and visualization: Matplotlib - Basic plot, Ticks, Labels, and Legends. (05) Working with CSV files. Reading, Manipulating, and Processing Data. Working with Excel Document: Installing the openpyxl Module, Reading Excel Documents, Project: Reading Data from a Spreadsheet, Writing Excel Documents, Project: Updating a Spreadsheet, Setting the Font Style of Cells, Font Objects, Formulas, Adjusting Rows and Columns, Charts.(07)

Lectures: 12

Total Lectures: 45

SEMESTER – III

Paper No.	**303**
Paper Name	**Computer Organization and Architecture**

UNIT-I

Register Transfer Language , Register Transfer, Bus and memory Transfer,(03) Micro-Operations, Arithmetic Micro-Operation, Logic Micro-Operation, Shift Micro-Operation, Common Bus Organization ,Multiple Bus Organization, Arithmetic Logic Shift UNIT (03), Addition and Subtraction Algorithm(02) ,Multiplication Algorithm (Booth Multiplication Algorithm) (03)

Lectures: 11

UNIT-II

Instruction Codes , Computer Instruction(02) , Computer Registers (DR, AR, AC, IR, PC, TR, INPR, OUTR) , Instruction Cycle (Fetch and Decode) (03), Control Unit Organization: Functional Requirements of a Control UNIT, Hardwired Control Unit(03) , Micro programmed Control UNIT (Microinstructions , Micro program Sequencer) (04)

Lectures:12

UNIT-III

Processor Design: General Register Organization , Stack Organization(Push and Pop Operation , Register Stack , Memory Stack) (03), Instruction Format(Zero Address Instructions , One Address Instruction ,Two address Instruction , Three Address Instruction) ,Data transfer and manipulation(Data transfer Instructions , Data Manipulation Instructions , Program Control Instructions) (03) ,RISC and CISC Architecture (02) , Addressing Schemes: Immediate Addressing, Direct Addressing, Indirect Addressing, Register Addressing, Register Indirect Addressing, Displacement Addressing (04)

Lectures: 12

UNIT-IV

Input-Output Organization: I/O Interface, Modes of transfer, Interrupts & Interrupt handling, Direct Memory access (DMA controller , DMA transfer)(04) ,**Memory Organization:** Memory Hierarchy, Main Memory (RAM and ROM Chips)(02), Auxiliary memory(Magnetic disks , magnetic Tapes), Cache memory (Hit Ratio , mapping techniques :Associative mapping ,Direct mapping ,Set associative mapping), Introduction to Virtual Memory(04).

Lectures:10

Total Lectures:45

SEMESTER – III

Paper No.	**304**
Paper Name	**Designing and Analysis of Algorithms**

UNIT – I

Mathematical Preliminaries: Review of growth functions, (03) Introduction of algorithm, characteristics of algorithm, (03) Complexity of algorithm, Efficiency of Algorithm (02), Asymptotic notations. (03)

Lectures: 11

UNIT – II

Sorting and Searching- Insertion sort, Bubble sort, Selection sort, (04) Quick sort, Merge sort, (02) Radix Sort, Bucket Sort, (02) External Sorting (01) Searching-Sequential and Binary search and their complexities. (03)

Lectures: 12

UNIT -III

Greedy algorithms, general characteristics of greedy algorithms, (03) job sequencing, minimum spanning tree, Single source shortest paths. (04) The knapsack problem, task scheduling problem, (01) Divide and Conquer Technique: merge sort, quick sort, strassen's matrix multiplication (04)

Lectures: 12

UNIT - IV

Dynamic Programming: General method Matrix multiplications, Travelling sales person problem, binary search trees, 0/1 knapsack problem (04) **Backtracking** – n-Queen's Problem, Hamiltonian Circuit problem, graph colouring (03) **Branch and bound** –Traveling salesman problem. FIFO branch and bound (03)

Lectures: 10

Total Lectures: 45

SEMESTER – III

Paper No.	**305**
Paper Name	**Organizational Behavior**

UNIT -I

Introduction to organizational behavior: nature, scope, challenges of organizational behavior, managing people and organization, (06) organizational behavior in global context organizational behavior model.(05)

Lectures:11

UNIT- II

Work teams - concept and types. Team effectiveness. Team creation. Power-bases of power, contingency approach to power.(06) Politics -concept, causes and management, leadershipconcept, styles, theories- Trait theory, charismatic leadership theory, behavioral theory (06)

Lectures:12

UNIT- III

Organization theory - classical, neoclassical and modern. Organizational culture concept, impact, creating organization culture. (06) Organizational design-importance types of OD, consequences of poor OD (06)

Lectures:12

UNIT -IV

Organizational change-factors, planned change, change resistance to change and overcoming resistance, Organization development- definition, process of organizational development. (05) Action research model, Organizational development interventions-sensitivity training and organizational development .(05)

Lectures:10

Total Lectures:45

SEMESTER – III

Paper No.	**306**
Paper Name	**Practical (DBMS, Python)**

Bachelor of Computer Application (BCA)

NEP

SEMESTER – III

[Minor Paper]

Paper No.	**MNRCMP031**
Paper Name	**Principles of Organisational Theories [Minor]**

UNIT-I

Introduction to organizational behaviour: nature, scope, challenges of organizational behaviour (05), managing people and organization (05), organizational behaviour in global context organizational behaviour model (05).

Lectures: 15

UNIT-II

Work teams - concept and types. Team effectiveness. Team creation. Power-bases of power, contingency approach to power (05). Politics -concept, causes and management, leadership concept, styles, (05) theories- Trait theory, charismatic leadership theory, behavioural theory (05).

Lectures: 15

UNIT-III

Organization theory - classical, neoclassical and modern (05). Organizational culture concept impact, creating organization culture (05). Organizational design-importance types of OD, consequences of poor OD (05).

Lectures: 15

UNIT-IV

Organizational change-factors, planned change, change resistance to change and overcoming resistance (05), Organization development- definition, process of organizational development. (05) Action research model, Organizational development interventions-sensitivity training and organizational development (05).

Lectures: 15

Total Lectures: 60

SEMESTER – III

[Vocational Paper]

Paper No.	**VOCCMP031**
Paper Name	**ICT Tools and its Applications**

SEMESTER – IV

Paper No.	**401**
Paper Name	**Cloud Computing**

UNIT – I

Introduction to Cloud Computing: Definition of Cloud – Evolution of Cloud Computing, Underlying Principles of Parallel and Distributed Computing, (05) Cloud Characteristics – Elasticity in Cloud, On-demand Provisioning. (05)

Lectures: 10

UNIT- II

Cloud Enabling Technologies Service Oriented Architecture: REST and Systems of Systems, Web Services – Publish, Subscribe Model – Basics of Virtualization – Types of Virtualization – (06) Implementation Levels of Virtualization – Virtualization Structures – Tools and Mechanisms – Virtualization of CPU – Memory – I/O Devices –Virtualization Support and Disaster Recovery.(05)

Lectures: 11

UNIT -III

Cloud Architecture, Services and Storage: Layered Cloud Architecture Design – NIST Cloud Computing Reference Architecture – Public, Private and Hybrid Clouds – IaaS – PaaS – SaaS – (06) Architectural Design Challenges – Cloud Storage – Storage-as-a-Service – Advantages of Cloud Storage – Cloud Storage Providers – S3. (06)

Lectures: 12

UNIT -IV

Resource Management and Security in Cloud: Inter Cloud Resource Management, Resource Provisioning and Resource Provisioning Methods, Global Exchange of Cloud Resources, Security (06) Overview – Cloud Security Challenges, Software-as-a-Service Security, Security Governance, Virtual Machine Security – IAM – Security Standards.(06)

Lectures: 12

Total Lectures: 45

SEMESTER – IV

Paper No.	**402**
Paper Name	**Java Programming**

UNIT -I

Java programming language overview, History of Java, features of Java, Java architecture,(04) referring to applets and applications, compiling and running a program, Byte code and JVM, (03) Java primitives data types, Variables: Declaration, Initialization, scope and lifetime, Command line arguments. (04)

Lectures**: 11**

UNIT -II

Java Operators, Typecasting, (02) Control statements and looping structure: if statement, switch construct, while loop, the for loop, the do loop, the break statement, the continue statement, retun statements, arrays. (03) Classes and object: concept of OOPS, constructors, types of constructors ,garbage collector, finalize(),(03) java method and object declaration, java methods, types of methods, passing arguments, methods overloading, constructor overloading, access specifiers. (04)

Lectures: 12

UNIT -III

Inheritance, using inheritance,this and super keywords, (04) overriding methods, Abstract classes, Package and Interface: defining packages, importing packages, grouping classes in packages, creating Interface, implementing interface, using interface ,extending interface. (04) String handling: using the string class as data type, using strings, string constructors, using string without the new modifier, using string methods, StringBuffer class. Vector, exception handling, multithreading (04)

Lectures: 12

UNIT -IV

Graphical user interface: defining applet, applet lifecycle, applet tags, applet methods, drawing lines, rectangles, polygons (04) AWT package class hierarchy, AWT controls, button, labels, text field, text area, check box, radio box, list, scroll bars, choice, frame, adding a button, creating panels ,layout managers. (04) Introduction to Servlets and RMI(03).

Lectures: 11

Total Lectures: 45

SEMESTER – IV

Paper No.	**403**
Paper Name	**Software Engineering**

UNIT -I

Software Engineering Fundamentals**: Definition** of software product and process(03), Software Crisis, Software development paradigms(03), Software lifecycle models: Waterfall Model, Prototyping Model, Iterative Enhancement Model, Evolutionary Development Model and Spiral Model(04).

Lectures: 10

UNIT -II

Software Requirement Analysis & Specification: System specification, Software requirements specification (SRS) standards (02), Formal specification methods, Specification tools, Requirements validation and management (02). Software Process: Software Process and Models, Tools and Techniques of Process Modeling, Product and Process(04). important qualities of software product and process: correctness, reliability, robustness, user friendliness, verifiability, maintainability, reusability, portability, data abstraction, modularity, Principles of software engineering(04).

Lectures: 12

UNIT -III

Software Design: Software architecture, Modular design - cohesion and coupling (02), Process-oriented design, Data-oriented design, User-interface design, Real-time software design(04). CASE Tools: Relevance of CASE tools, High-end and low-end CASE tools(02), Automated support for data dictionaries, DFD, ER diagrams(04).

Lectures: 12

UNIT -IV

Coding and Testing: Choice of Programming languages, Coding standards(02), Introduction to Testing Process, Functional & Structural Testing(02), Testing Activities like UNIT, Integration & System Testing(04), Testing tools and workbenches(03).

Lectures: 11

Total Lectures: 45

SEMESTER – IV

Paper No.	**404**
Paper Name	**Web Designing**

UNIT -I

Internet: Basic Internet protocols, Client Server Architecture – Two-Tier, Multi-Tier, DNS system, URL, Search Engine (05) World Wide Web, understanding basics of HTML tools, entering tags and attributes, applying structural tags, Lists, linking documents: URL, types of URL's, constructing link anchors, inserting e-mail link, inserting images, using images as links, using background images, Image Map. (05)

Lectures: 10

UNIT –II

Tables: creating basic tables, spanning rows and columns, adding captions, formatting tables, adding and formatting borders, Frames: creating frames, accommodating non-framed browsers, HTML forms, Form tools, HTML 5 Form Tools, Various Forms Designing (05) Style sheets: CSS Background, CSS Font, CSS Text, CSS Box Model, CSS position, CSS Pseudo class, CSS layers with Z-index, CSS Float and CSS display properties (07)

Lectures: 12

UNIT III

JavaScript: what is javascript, adding javascript, datatypes in javascript, dialog boxes, control statements: decision control statements and loops. (05) Javascript Arrays: Declariong Arrays, Accessing array. Defining Functions, calling functions, local and global variables, frequently used events, adding event handlers, String Module, date time module, validating html forms (07)

Lectures: 12

UNIT IV

Introduction to Bootstrap & Setup, Typography & Utilities, Bootstrap Grid System, Containers & Breakpoints, Section, Buttons & Button Groups, (05) Navbar & Navs, List Groups and Badges, Forms, Inputs & Input Groups, Alerts, Progress Bars, Working with Tables, Cards, Media objects & Jumbotron, Responsive Tables & Images.(06)

Lectures: 11

Total Lectures: 45

SEMESTER – IV

Paper No.	**405**
Paper Name	**Computer Oriented Numerical Methods**

UNIT -I

Numerical system and error and types of error, Floating point Arithmetic, Source of error, Zeros of transcendental equations and polynomials (03), system of non linear equation(04), Solution of Algebraic and transcendental function- Bisection Method , Iteration Method, Method of false position, Newton Raphson method, Generalized Newton's method (04).

Lectures: 11

UNIT -II

Solution of system of linear equation, Gaussian elimination method, Gauss Jordon method (04), Pivoting, Iterative methods of Jacobi and Gauss Seidel Methods (04), Matrix Inversion Method (02), Method of Factorization (02).

Lectures: 12

UNIT -III

Interpolation, Errors in Polynomial Interpolation, Finite Differences, Forward differences, Backward Differences, Central Differences (02), Newton's formula for interpolation, Guass's central difference formula (03), Stirling's Formula, Bessel's formula (03), Lagrange's Interpolation Formula, Error in Lagrange's Interpolation Formula (04).

Lectures: 12

UNIT -IV

Numerical differentiation, Error in Numerical differentiation , Cubic Spline method (03), Numerical Integration: Trapezoidal rule, Simpson's 1/3 rule, Simpson's 3/8 rule , Romberg's Interpolation (04), Numerical solutions of ordinary differential equations: Solution by Taylor's series, Euler's Method, Runge-kutta Methods (03).

Lectures: 10

Total Lectures: 45

SEMESTER – IV

Paper No.	**406**
Paper Name	**Practical (Java, HTML,CSS, JAVASCRIPT), Mini Project**

Semester – IV

[Vocational Paper]

Paper Code	**VOCCMP041**
Paper Name	**Introduction to Web Designing**

SEMESTER – V

Paper No.	**501**
Paper Name	**Programming and Problem Solving through C#**

UNIT -I

The .net Framework: introduction, common language runtime, common type system, common language specification, (05) the base class library, the .net class library, Intermediate language, (05) Just in time compilation, garbage collection, assemblies, web services, COM, localization(05)

Lectures: 15

UNIT -II

C# Basics: Introduction, data types, identifiers, variables, constants, C# statements, OOPs concept, (05) array and strings, operators, control statements, classes and objects(05), access modifiers, overloading, inheritance, overriding, interfaces. (05)

Lectures: 15

UNIT -III

Visual studio IDE features, introduction to Window forms, components, (03) control: textbox, label, linklabel, status bar, (03) checkedlistbox, combobox, listbox, listview, radiobutton, button, panel, (04) groupbox, dialog box, menu control, properties, methods, events of controls. (05)

Lectures: 15

UNIT -IV

ADO.net, the component model, creating database connection, database command, (03) data repeater, connecting to data sources, choosing a .net data provider, (03) manage a connection, building command objects, (04) executing commands, building datasets and datatables, data adapter (05)

Lectures: 15

Total Lectures: 60

SEMESTER – V

Paper No.	**502**
Paper Name	Mobile Application Development Using Flutter

Unit-I

Dart programming Basic: features, installation, basic syntax, comments ,keywords, data types, variables , Operators, **Control flow Statement:** if statement, if else, if elseif, switch, for,for..in, while,do while Loop.(10) **Function:** Dart function, Anonymous Function, main function ,Recursion(05)

Lectures : 15

Unit-II

Object Oriented concepts:-Class & object, constructor,this keyword,static keyword,super keyword,inheritance,super constructor,methods,overriding,getter and setters, Abstract class,Interface(08)

Dart Advance: Exception handling, typedef, metadata, collection, generics, packages, libraries, generators (07)

Lectures : 15

Unit-III

Flutter basics: Installation, Architecture, widgets, Layout ,gestures, state management ,flutter IDE(05)

Flutter widgets: Scaffold, container, row and column, Text, TextField,Buttons,Stack,Forms,Alert Dialog,(05) icons,images,card,Tabbar,Drawer,Lists,Gridview,Toast,checkbox,radio button,Progressbar, Slider, Navigation Bar(05)

Lectures : 15

Unit-IV

Android platform ,packages ,(05) splash screen google maps, REST API, Database ,Testing (10)

Lectures : 15

Total Lectures: 60

SEMESTER – V

Paper No.	**503**
Paper Name	**Artificial Intelligence and Data Analytics**

UNIT-I
Introduction, The importance of AI , Applications of AI , general issue in AI problem solving (02), production system (02), knowledge: Definition & importance, knowledge based systems , components of knowledge based system , intelligent agents (03) , introduction to fuzzy logic, genetic algorithm, basic concept of natural language processing (03).
Lectures: 10

UNIT-II
Search and Control Starategies, Examples of Search Problems(The eight puzzle, Travelling Salesman Problem) (03), Uninformed Search (Breadth First Search , Depth First Search , Depth First Iterative Depending Search , Bidirectional Search) and informed search (Heuristic Information , Hill Climbing Methods ,Best First Search , Branch and Bound Search ,A* Search) (04), Searching AND-OR Graphs, The AO* Algorithm , Game Playing : mini max search approach (04).
Lectures: 11

UNIT-III
Formalized Symbolic logics: Propositional logic , Syntax and Semantics for Propositional Logic ,Limitation of Propositional Logic (02), First order predicate Logic , Syntax and Semantics for FOPL ,Properties of Well Formed Formula (02), skolemisation , Conversion to Clausal Form , inference rules , unification, resolution principle (02), Introduction to ANN,CNN and different learning algorithms (06).
Lectures: 12

UNIT-IV
Semantic networks, frame system(03), value inheritance scripts(02), LISP and other AI programming languages(03), Introduction to Data Analytics: Sources and nature of data, classification of data (structured, semi-structured, unstructured), characteristics of data, introduction to Big Data platform, need of data analytics, evolution of analytic scalability, analytic process and tools, analysis vs reporting, modern data analytic tools, applications of data analytics.(04).
Lectures:12

Total Lectures: 45

SEMESTER – V

Paper No.	**504**
Paper Name	**Software Testing and Quality Management**

UNIT -I

Testing Software :Software Testing and its importance, Module Testing, Integration Testing, Top-down versus bottom up testing, Mixed testing.(03) Software Faults and Failures: Types of faults, Origins of Software Defects, The Cost of Repairing Defects(02)Verification and Validation :Verification Testing, Coding standards, Walk-Through, Formal Inspection(02), Verifying . Requirements, Verifying Functional Design, Validation Test Criteria, Design metrics. (03)

Lectures: 10

UNIT -II

Testing Techniques and Strategies: White-Box Testing: Flow graph notation, Cyclomatic Complexity, Control Structure and Loop Testing, Dataflow Testing.(04) Black-Box Testing: Graph-based testing methods, Equivalence partitioning, Boundary Value Analysis(03),UNIT Testing, Integration Testing, System Testing, Exhaustive Testing.(04)

Lectures: 11

UNIT -III

BuildingTests and Test Plans :Designing and Creating Tests, Maintaining Checklists, White-box Test Cases and Test Procedures, Test Data Selection and Outputs, Black-box test cases and test procedures, Planning and Creating Test Plans. (04) Testing Specialized Systems and Applications :Graphical User Interface (GUI) Testing, Usability Testing, Client/Server Architectures and Web Testing, Testing OO Systems, Volume and Stress Testing (04) Testing Measurements: Software Size and Complexity, Function Point Analysis(04)

Lectures: 12

UNIT -IV

Quality Assurance and Standards:Quality and Quality Assurance (QA), Techniques of quality Assurance, Software Testing and QA,(04)Software Development Models, Quality metrics(03), Configuration Management, Quality management models *(ISO,* SPICE, IEEE, and CMM).(05)

Lectures: 12

Total Lectures: 45

SEMESTER – V

Paper No.	**505**
Paper Name	**Computer Graphics**

UNIT -I

Application of computer graphics, graphics devices- LED,LCD,CDA,FPD (04), drawing geometry, Line drawing Algorithms: DDA and Bresenham, Functions implementation(03), Cathode Ray Tube implementation.(03)
Lectures: 10

UNIT -II

2D transformation: Translation, Rotation, Scaling, Reflection, Shearing(02), Circle Drawing: Bresenham and Mid-point sub divison function implementation(02), Clipping: End Point Codes, Cohen Sutherland, Mid point Subdivision Algorithm(04), Mapping, dragging,echoing, Polygon Filling, character generation(04)
Lectures: 12

UNIT -III
3D graphics transformation-translation,rotation,scaling,reflection,shearing(02), Projection:parallel, Prespective projection(05), Hidden surface, removal algorithm method, back face removal algorithm,Z Buffer Algorithm, floating horizon technique(05)
Lectures: 12

UNIT -IV
Tweening-interpolation, morphing technique(03), GKS- primitive,work station(03), multimedia application-animation principle,animation tools(05)
Lectures: 11
Total Lectures:45

SEMESTER – V

Paper No.	**506**
Paper Name	**Practical (C#)**

SEMESTER – VI

Paper No.	**601**
Paper Name	**Data Warehousing and Mining**

UNIT- I

Data Warehouse fundamentals: The Data Warehouse –Introduction, characteristics, its competitive advantages, (05) Operational Database Systems and Data Warehouse (OLTP & OLAP). (05)Multidimensional Data Models: Types of Data, from Tables and Spreadsheets to Data Cubes (05)

Lectures: 15

UNIT -II

Principles of dimensional modeling : Identifying Facts and Dimensions, Designing Fact Tables, Designing Dimension Table, Data Warehouse Schemas, OLAP Operations, Data Extraction, Cleanup & Transformation, Star, snowflake and galaxy schemas for multidimensional databases.(08) Architecture for a warehouse, Steps for construction of Data Warehouses, Data Marts, Metadata. Different OLAP operations, OLAP Server: ROLAP, MOLAP and HOLAP (07)

Lectures: 15

UNIT -III

Data Mining and its Functionalities: From Data warehousing to data mining, Motivation, Knowledge Discovery Process, objectives of Data Mining, the business context for DM, Process improvement, marketing and CRM, (10) Tools of Data Mining**.** Data preparation, Data Mining Techniques: Statistical techniques, Characterization and discrimination, Association and market basket analysis, Classification and Prediction, Cluster analysis, Outlier analysis.(10)

Lectures: 20

UNIT –IV

Data Mining Applications : Text Mining, Spatial Databases, Web Mining. Case studies in building business environment. ,(05) Applications in telecommunications industry, retail, target marketing, fraud protection, health care, science, ecommerce, banking and finance.(05)

Lectures: 10

Total Lectures: 60

SEMESTER – VI

Paper No.	**602**
Paper Name	**Linux & Shell Programming**

UNIT - I

Linux introduction and file system – Basic Features, Advantage, Installing requirement , Basic Architecture of Unix/Linux System, Kernel , Shell(05). Linux File System-Boot block, super block, Inode Table, data blocks, how Linux Access Files, storage files, Linux Standard directories(05). Commands for files and directories cd , ls, cp ,md, rm , mkdir , rmdi , pwd , file ,more, less, creating and viewing files using cat , file comparison- cmp & comm., view files, disk related commands, checking disk free spaces(05).

Lectures: 15

UNIT- II

Essential Linux commands, Understanding shells, Processes in linux- process fundamentals, connecting process with pipes, tee, Redirecting input output, manual help(05). Background processing ,managing multiple process, changing process priority with nice, scheduling of process at command, cron , batch commands ,kill ps, who, sleep, Printing commands find ,sort, touch , file ,file related commands ws, sat, cut , dd , etc.(05) Mathematical commands- bc , expr , factor , UNITs (02). Creating & editing files with vi, joe & vim editor (03).

Lectures: 15

UNIT- III

System Administration: Common Administrative tasks, identifying administrative files, configuration and log files, role of system administrator, managing user accounts- adding and deleting users, changing permission and ownership, creating and managing groups, modifying group attributes, temporary disable user accounts, creating and monitoringfile system(07) , checking and monitoring system performance, file security and permission, becoming super user using su. Getting system information with uname, host name, disk partition & sizes, users, kernel(05). Backup and restore files, reconfiguration hardware with kudzu, installing and removing packages with rpm command(03)

Lectures: 15

UNIT- IV

Shell programming- Basic of shell programming, various types of shells available in Linux, comparisons between various shells, shell programming in bash, read command, conditional and looping statements,(05) case statements, parameter passing and arguments, Shell variables, system shell variables, shell keywords, Creating Shell programs for automate system tasks(05). Simple filter commands- pr, head, tail, cut, paste, sort, uniq, tr. Filter using regular expressions- grep, egrep and sed. Awk programming- report printing with awk(05).

Lectures: 15

Total Lectures- 60

SEMESTER – VI

Paper No.	**603**
Paper Name	**Cryptography and Network Security**

UNIT- I

Introduction to Network Security, Aspects of Network Security, Need For Security, Goals Of Network Security, (05) Features of a good Security policy, security attacks, security services and mechanisms, network security,(05) Firewall: type, hardware and software, design principles (05)

Lectures: 15

UNIT- II

Encryption/decryption, conventional encryption model, (05) conventional encryption algorithms: character level encryption, bit-level encryption, private key encryption, (05) Data encryption Standard algorithm, strength of DES (05)

Lectures: 15

UNIT- III

Public key encryption, principles of public key cryptography systems, RSA algorithm, security of RSA,(05) Fermet's and euler's theorem, primality, The chineese remainder theorem, (05) Digital signature, application for public key cryptosystems (05)

Lectures: 15

UNIT- IV

E-mail security, PGP, S\MIME security, S\MIME functionality, cryptographic algorithms,(04) IP Security: application of IPSec, Benefits of IPSec, IPSec architecture, IPSec Services, Authentication header, Encapsulating Security payload. (07) Web Security: threats on web, SSL and Transport layer Security. System security : virus and Intruders.(04)

Lectures: 15

Total Lectures: 60

SEMESTER – VI

Paper No.	**604**
Paper Name	**Industrial Training and Project**